

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
```

```
def predict(X, weights, lambd=1):
    net = np.dot(X, weights)
    return (1 / (1 + np.exp(-lambd * net)))

def calculate_dw(X, y, y_pred):
    return ((y_pred - y) * (y_pred) * (1 - y_pred) * X)

def calculate_error(yi, y_pred):
    return np.square(y_pred - yi)
```

```
def train_vanilla(X, y, weights, epochs=500, lr=0.001, gamma=0.9):
    vel = 0
    for epoch in range(epochs):
        dw = 0
        error = 0
        w_lookahead = weights - (gamma * vel)

        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, w_lookahead)
            dw += calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)

        vel = (gamma * vel) + (lr * dw)
        weights -= vel
        error /= (2 * len(X))

        if (epoch + 1) % 50 == 0:
            print(f"Weights after epoch {epoch + 1}: ", weights)
            print(f"Error after epoch {epoch + 1}: ", error)
```

```
def train_stochastic(X, y, weights, epochs=500, lr=0.001, gamma=0.9):
    vel = 0
    for epoch in range(epochs):
        error = 0

        for Xi, yi in zip(X, y):
            w_lookahead = weights - (gamma * vel)
            y_pred = predict(Xi, w_lookahead)
            dw = calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)
            vel = (gamma * vel) + (lr * dw)
            weights -= vel

        error /= (2 * len(X))

        if (epoch + 1) % 50 == 0:
            print(f"Weights after epoch {epoch + 1}: ", weights)
            print(f"Error after epoch {epoch + 1}: ", error)
```

```
def train_mini_batch(X, y, weights, epochs=500, lr=0.001, gamma=0.9, bs=32):
    vel = 0
    for epoch in range(epochs):
        dw = 0
        error = 0
        i = 0
        w_lookahead = weights - (gamma * vel)

        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, w_lookahead)
            dw += calculate_dw(Xi, yi, y_pred)
            i += 1
            error += calculate_error(yi, y_pred)

            if i % bs == 0 or i == len(X):
                vel = (gamma * vel) + (lr * dw)
                weights -= vel
                dw = 0
                w_lookahead = weights - (gamma * vel)

        error /= (2 * len(X))

        if (epoch + 1) % 50 == 0:
            print(f"Weights after epoch {epoch + 1}: ", weights)
            print(f"Error after epoch {epoch + 1}: ", error)
```

```
df = pd.read_csv('bank_note.csv')
df.insert(4, 'x0', 1)
```

```
X = df[['variance', 'skewness', 'curtosis', 'entropy', 'x0']].values
y = df['class']
```

```
X_train, y_train, X_test, y_test = train_test_split(X, y, random_state=13, test_size=0.2)
```

```
weights = input('Enter 4 weights and 1 bias: ').split()
weights = np.array([float(weight) for weight in weights], dtype='longdouble')
```

```
print('Nesterov Accelerator GD (Vanilla)')
train_vanilla(X, y, weights.copy())
```

```
➡ Nesterov Accelerator GD (Vanilla)
Weights after epoch 50: [-3.72513351 -2.23946531 -2.63124085 -0.64636901  2.68248116]
Error after epoch 50:  0.004082919528475229
Weights after epoch 100: [-3.70683594 -2.06163838 -2.53118121 -0.32000033  3.39726197]
Error after epoch 100:  0.003236919062077093
Weights after epoch 150: [-3.75047664 -2.05563292 -2.54339637 -0.2500227  3.57421538]
Error after epoch 150:  0.0031861826133179312
Weights after epoch 200: [-3.81428895 -2.08400725 -2.58355819 -0.24260628  3.65796628]
Error after epoch 200:  0.0031667937587036973
Weights after epoch 250: [-3.8816543 -2.11606703 -2.62656328 -0.24485354  3.71976416]
Error after epoch 250:  0.003150571844462434
Weights after epoch 300: [-3.94678231 -2.1476307 -2.66840945 -0.24911844  3.77462285]
Error after epoch 300:  0.0031361003778591448
Weights after epoch 350: [-4.0088642 -2.17785476 -2.70837022 -0.25364241  3.82587144]
Error after epoch 350:  0.0031230752071045153
Weights after epoch 400: [-4.06795299 -2.20666477 -2.74643407 -0.25804637  3.87445563]
Error after epoch 400:  0.0031112902512163065
Weights after epoch 450: [-4.12426459 -2.23414122 -2.78272806 -0.26225675  3.92075955]
Error after epoch 450:  0.0031005790737727895
Weights after epoch 500: [-4.17802703 -2.26038773 -2.81739453 -0.26626944  3.96501348]
Error after epoch 500:  0.0030908041894093176
```

```
print('Nesterov Accelerator GD (Stochastic)')
train_stochastic(X, y, weights.copy())
```

```
➡ Nesterov Accelerator GD (Stochastic)
Weights after epoch 50: [-1.78832473 -1.09499963 -1.26125843 -0.14033199  2.02572752]
Error after epoch 50:  0.004760527334860335
Weights after epoch 100: [-2.17715925 -1.30631838 -1.52699482 -0.1494569  2.39643232]
Error after epoch 100:  0.004164190815010794
Weights after epoch 150: [-2.44794548 -1.44111437 -1.70270582 -0.16200781  2.62019199]
Error after epoch 150:  0.0039183583116630194
Weights after epoch 200: [-2.66097815 -1.54313988 -1.83813856 -0.17274211  2.78932972]
Error after epoch 200:  0.003772946841441464
Weights after epoch 250: [-2.83878806 -1.62685964 -1.95026783 -0.18186525  2.92894738]
Error after epoch 250:  0.003672996610109214
Weights after epoch 300: [-2.99251485 -1.69872018 -2.04693725 -0.18980778  3.04950268]
Error after epoch 300:  0.0035984529795541277
Weights after epoch 350: [-3.12855848 -1.76215124 -2.13244623 -0.19687422  3.15643951]
Error after epoch 350:  0.0035399521108837755
Weights after epoch 400: [-3.2509671 -1.81920857 -2.20943049 -0.20326677  3.25300568]
Error after epoch 400:  0.0034924180692374436
Weights after epoch 450: [-3.36248069 -1.8712327 -2.27963754 -0.20912309  3.34132333]
Error after epoch 450:  0.0034528087567235943
Weights after epoch 500: [-3.46504975 -1.91915347 -2.34429458 -0.21454059  3.42287141]
Error after epoch 500:  0.0034191647822317196
```

```
print('Nesterov Accelerator GD (Mini Batch)')
train_mini_batch(X, y, weights.copy())
```

```
➡ Nesterov Accelerator GD (Mini Batch)
Weights after epoch 50: [-1.84011053 -1.10843593 -1.2627129 -0.12812667  2.0187505 ]
Error after epoch 50:  0.0048066605413041
Weights after epoch 100: [-2.22288091 -1.30657444 -1.5228339 -0.13360726  2.389217 ]
Error after epoch 100:  0.004204829115502439
Weights after epoch 150: [-2.49111634 -1.43863996 -1.69813733 -0.14675294  2.61163658]
Error after epoch 150:  0.003948123142184919
Weights after epoch 200: [-2.70236952 -1.54007112 -1.83385998 -0.15871609  2.77974934]
Error after epoch 200:  0.0037950142843448044
Weights after epoch 250: [-2.87867019 -1.62377356 -1.94631142 -0.16913731  2.91847547]
Error after epoch 250:  0.003689725184325798
Weights after epoch 300: [-3.03099532 -1.6957692 -2.04320772 -0.1783176  3.03816071]
Error after epoch 300:  0.0036113548987655045
Weights after epoch 350: [-3.1656769 -1.7593471 -2.12882964 -0.18652829  3.14420443]
Error after epoch 350:  0.0035500126203688092
Weights after epoch 400: [-3.28673202 -1.81651212 -2.20581969 -0.19396604  3.2398401 ]
Error after epoch 400:  0.003500301918335255
Weights after epoch 450: [-3.39688641 -1.86858717 -2.27593788 -0.20077234  3.32718801]
Error after epoch 450:  0.0034589800029814284
```

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Weights after epoch 500: [-3.49808467 -1.91649815 -2.34042416 -0.20705164  3.4077303 ]  
Error after epoch 500:  0.003423957227593001
```